# Architectural Paradigms in Flexible Media

Understanding when to dictate (Art Direction) versus when to empower (Resolution Switching)

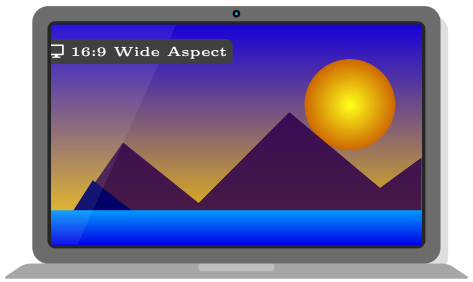
## Art Direction (<picture>)

art-direction.html

```html
<picture>
  <source media="(min-width: 800px)" srcset="wide-crop.jpg">
  <img src="square-crop.jpg">
</picture>
```

✔ Viewport ≥ 800px

↰ Fallback (< 800px)




Desktop (≥800px): `wide-crop.jpg`


Mobile (<800px): `square-crop.jpg`

↔ Visual Art Direction: Aspect ratio & framing physically altered

### Browser is DICTATED To

- **Use Case:** The image content physically changes (cropping, aspect ratio, focal point) at different breakpoints.
- **Constraint:** The browser *must* obey the media query strictly. It has no freedom to optimize based on network speed or device density.

## Resolution Switching (srcset+sizes)

resolution-switching.html

```html
<img
  srcset="img-400.jpg 400w,
       img-800.jpg 800w"
  sizes="(min-width: 800px) 50vw, 100vw">
```

**PRE-CSS PARSE PHASE: Browser Preload Scanner Running**

### srcset Descriptor Map

Available files & intrinsic widths:

`img-400.jpg` $\rightarrow W_{\text{available}} = 400\text{w}$

`img-800.jpg` $\rightarrow W_{\text{available}} = 800\text{w}$

### sizes Layout Budget

Informs rendered slot width before CSS:

$$W_{\text{layout}} = \begin{cases} 50\text{vw} & (\geq 800\text{px}) \\ 100\text{vw} & (\text{default}) \end{cases}$$

*(Eliminates pre-CSS blindness)*

### Browser Selection Engine: Mathematical Evaluation

$$\text{Effective DPR} = \frac{W_{\text{available}}}{W_{\text{layout}}} \implies \text{Target Width} = W_{\text{layout}} \times \text{Device DPR}$$

### WITHOUT sizes (Blindness)

- CSS stylesheet not parsed yet.
- Browser knows $W_{\text{available}} \in \{400, 800\}$, but rendered slot $W_{\text{layout}} =$ UNKNOWN.

$$\frac{W_{\text{available}}}{\textbf{? (Unknown)}} = \text{Cannot Calc Density}$$

$\implies$ Preload scanner is blinded; must fall back to 100vw or wait for CSS.

### WITH sizes (Empowered)

- $W_{\text{layout}}$ supplied directly in HTML.
- Example: Viewport=800px $\implies W_{\text{layout}} =$ 400px.
  Device DPR = 2.0× (Retina).

$$\frac{800\text{w}}{400\text{px}} = \mathbf{2.0}\times \implies \text{Exact Match}$$

$\implies$ Instantly downloads `img-800.jpg` with zero preload delay!

### Browser is EMPOWERED

- **Use Case:** The image content is identical, but scalable graphics are needed to support various layout sizes and high-DPI screens (Retina).
- **Advantage:** By supplying $W_{\text{layout}}$ (`sizes`) and $W_{\text{img}}$ (`srcset`), the browser heuristically picks the best image, factoring in device pixel ratio and user bandwidth.